

Casos de Uso — Código PlantUML

Sistema de Evaluación de Riesgo Crediticio para Inmobiliaria

Cómo usarlo: Copie cada bloque de código (desde @startuml hasta @enduml) y péguelo en el editor de PlantUML (<https://www.plantuml.com/plantuml>) o en la extensión de PlantUML de VS Code.

Criterio aplicado (include vs extend):

<<include>> = comportamiento OBLIGATORIO, siempre se ejecuta (el CU está incompleto sin él).

<<extend>> = comportamiento OPCIONAL o CONDICIONAL, solo se dispara bajo ciertas condiciones.

CU-01 — Iniciar sesión (con 2FA)

Actores: Propietario, Administrador, Operador · **Dependencias:** RF-07, RNF-03, RNF-04, RNF-05

```
@startuml
left to right direction

actor Propietario as PROP
actor Administrador as ADMIN
actor Operador as OPER

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Iniciar sesion" as CU01
    usecase "Validar credenciales" as V1
    usecase "Solicitar codigo 2FA (OTP)" as V2
    usecase "Denegar acceso" as V3
}

PROP --> CU01
ADMIN --> CU01
OPER --> CU01

CU01 ..> V1 : <>
V2 ..> CU01 : <>
V3 ..> CU01 : <>

note bottom of V2
    Se dispara solo si la cuenta
    tiene 2FA activo.
end note

note bottom of V3
    Se dispara tras varios
    intentos fallidos.
end note

@enduml
```

CU-02 — Gestionar usuarios

Actores: Administrador, Propietario · **Dependencias:** RF-10, RF-08, RNF-03, RNF-10

```
@startuml
left to right direction

actor Administrador as ADMIN
actor Propietario as PROP

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Gestionar usuarios" as CU02
    usecase "Iniciar sesion" as CU01
    usecase "Registrar usuario" as U1
    usecase "Editar usuario" as U2
    usecase "Activar / Desactivar usuario" as U3
    usecase "Registrar en auditoria" as AUD
}

ADMIN --> CU02
PROP --> CU02

CU02 ..> CU01 : <>
CU02 ..> AUD : <>
U1 ..> CU02 : <>
U2 ..> CU02 : <>
U3 ..> CU02 : <>

@enduml
```

CU-03 — Gestionar clientes

Actores: Operador, Administrador · **Dependencias:** RF-01, RF-02, RNF-04, RNF-05, RNF-10

```
@startuml
left to right direction

actor Operador as OPER
actor Administrador as ADMIN

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Gestionar clientes" as CU03
    usecase "Iniciar sesion" as CU01
    usecase "Validar duplicidad de documento" as VAL
    usecase "Registrar cliente" as C1
    usecase "Editar cliente" as C2
    usecase "Ver detalle de cliente" as C3
    usecase "Eliminar cliente (baja logica)" as C4
}

OPER --> CU03
ADMIN --> CU03

CU03 ..> CU01 : <>
CU03 ..> VAL : <>
C1 ..> CU03 : <>
C2 ..> CU03 : <>
C3 ..> CU03 : <>
C4 ..> CU03 : <>

@enduml
```

CU-04 — Configurar motor de reglas y criterios

Actores: Administrador · **Dependencias:** RF-03 (config), RF-12, RNF-03, RNF-08, RNF-10

```
@startuml
left to right direction

actor Administrador as ADMIN

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Configurar motor de reglas" as CU04
    usecase "Iniciar sesion" as CU01
    usecase "Definir reglas IF-THEN" as R1
    usecase "Asignar pesos y umbrales" as R2
    usecase "Configurar criterios ponderados" as R3
}

ADMIN --> CU04

CU04 ..> CU01 : <>
CU04 ..> R1 : <>
CU04 ..> R2 : <>
R3 ..> CU04 : <>

@enduml
```

CU-05 — Evaluar riesgo de cliente (caso de uso central)

Actores: Operador · **Dependencias:** RF-03, RF-04, RF-05, RNF-01, RNF-03, RNF-10

```
@startuml
left to right direction

actor Operador as OPER

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Evaluar riesgo de cliente" as CU05
    usecase "Iniciar sesion" as CU01
    usecase "Verificar lista de sanciones (ONU/OFAC)" as LN
    usecase "Aplicar reglas y calcular puntaje" as E1
    usecase "Clasificar riesgo" as E2
    usecase "Generar reporte de la evaluacion" as E3
}

OPER --> CU05

CU05 ..> CU01 : <>
CU05 ..> LN : <>
CU05 ..> E1 : <>
CU05 ..> E2 : <>
E3 ..> CU05 : <>

note bottom of LN
    Si el cliente coincide con la
    lista, el riesgo se fuerza a ALTO.
end note

@enduml
```

CU-06 — Generar reportes

Actores: Operador, Administrador · **Dependencias:** RF-06, RNF-01, RNF-04, RNF-09

```
@startuml
left to right direction

actor Operador as OPER
actor Administrador as ADMIN

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Generar reportes" as CU06
    usecase "Iniciar sesion" as CU01
    usecase "Consultar evaluaciones" as G1
    usecase "Exportar a PDF" as G2
    usecase "Ver tablero de estadisticas" as G3
}

OPER --> CU06
ADMIN --> CU06

CU06 ..> CU01 : <>
CU06 ..> G1 : <>
G2 ..> CU06 : <>
G3 ..> CU06 : <>

@enduml
```

CU-07 — Consultar auditoría / historial

Actores: Administrador, Propietario · **Dependencias:** RF-09, RNF-10, RNF-03, RNF-04

```
@startuml
left to right direction

actor Administrador as ADMIN
actor Propietario as PROP

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Consultar auditoria" as CU07
    usecase "Iniciar sesion" as CU01
    usecase "Filtrar bitacora" as A1
    usecase "Ver detalle de evento" as A2
}

ADMIN --> CU07
PROP --> CU07

CU07 ..> CU01 : <>
A1 ..> CU07 : <>
A2 ..> CU07 : <>

note bottom of CU07
    El Administrador ve solo la auditoria
    de su empresa. El Propietario ve todas.
end note

@enduml
```

CU-08 — Gestionar empresas

Actores: Propietario · **Dependencias:** RF-11, RNF-03, RNF-04, RNF-08, RNF-10

```
@startuml
left to right direction

actor Propietario as PROP

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "Gestionar empresas" as CU08
    usecase "Iniciar sesion" as CU01
    usecase "Registrar empresa" as EM1
    usecase "Editar empresa" as EM2
    usecase "Activar / Desactivar empresa" as EM3
}

PROP --> CU08

CU08 ..> CU01 : <>
EM1 ..> CU08 : <>
EM2 ..> CU08 : <>
EM3 ..> CU08 : <>

@enduml
```

Diagrama general — Los 8 casos de uso

Actores: Propietario, Administrador, Operador

```

@startuml
left to right direction

actor Propietario as PROP
actor Administrador as ADMIN
actor Operador as OPER

rectangle "Sistema de Evaluacion de Riesgo Crediticio" {
    usecase "CU-01 Iniciar sesion (2FA)" as CU01
    usecase "CU-02 Gestionar usuarios" as CU02
    usecase "CU-03 Gestionar clientes" as CU03
    usecase "CU-04 Configurar reglas y criterios" as CU04
    usecase "CU-05 Evaluar riesgo de cliente" as CU05
    usecase "CU-06 Generar reportes" as CU06
    usecase "CU-07 Consultar auditoria" as CU07
    usecase "CU-08 Gestionar empresas" as CU08
    usecase "Verificar lista de sanciones" as LN
    usecase "Solicitar codigo 2FA (OTP)" as OTP
}

' Operador
OPER --> CU03
OPER --> CU05
OPER --> CU06

' Administrador
ADMIN --> CU02
ADMIN --> CU03
ADMIN --> CU04
ADMIN --> CU06
ADMIN --> CU07

' Propietario
PROP --> CU08
PROP --> CU07

' Include: autenticacion obligatoria
CU02 ..> CU01 : <>
CU03 ..> CU01 : <>
CU04 ..> CU01 : <>
CU05 ..> CU01 : <>
CU06 ..> CU01 : <>
CU07 ..> CU01 : <>
CU08 ..> CU01 : <>

' Include: evaluar riesgo verifica lista de sanciones (obligatorio)
CU05 ..> LN : <>

' Extend: 2FA es condicional al iniciar sesion
OTP ..> CU01 : <>

@enduml

```